LECTURE 03 • STUDY NOTES

Types of Computers & the Block Diagram

How computers are classified, and the functional plan they all share

BSc Computer Science (First Year) · Detailed notes with diagrams · 2-mark & 14-mark questions

Learning objectives. After studying these notes you should be able to:

- Classify computers by data handled, by purpose and by size.
- Distinguish analog, digital and hybrid computers.
- Compare super, mainframe, mini and micro computers.
- Draw and label the block diagram of a computer.
- Explain each functional unit and the difference between data flow and control flow.

1. Three ways of classifying computers

Computers are grouped in three different ways depending on what you are comparing:

- **By the data they handle:** Analog, Digital, Hybrid.
- **By purpose:** General-purpose, Special-purpose.
- **By size and power:** Supercomputer, Mainframe, Mini, Micro.

2. Classification by data: Analog, Digital, Hybrid

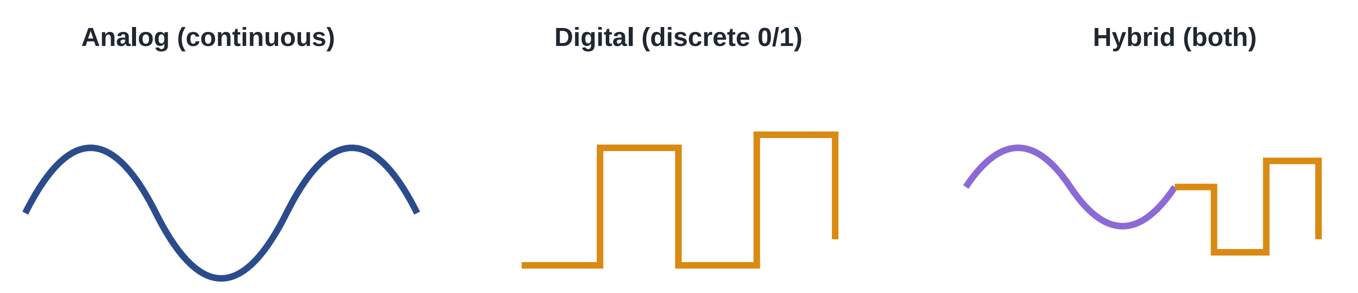


Fig 1. Analog signals are continuous; digital signals are discrete (0/1); hybrid uses both.

Type	Works with	Action	Examples
Analog	Continuous physical quantities (voltage, temperature)	Measures (approximate)	Speedometer, mercury thermometer, seismograph
Digital	Discrete data — binary 0 and 1	Counts (exact)	PC, laptop, smartphone, calculator
Hybrid	Both — analog input, digital processing	Measures then counts	ICU patient monitor, petrol pump

3. Classification by purpose

- **General-purpose computers** handle a wide range of tasks by loading different programs (e.g. a laptop or PC). Flexible, but not the fastest at any single job.
- **Special-purpose computers** are built and programmed for one specific task (e.g. a washing-machine controller, ATM, traffic-signal controller). They cannot be reprogrammed for other work, but perform their one job very efficiently.

4. Classification by size and power

Type	Power / Users	Typical use	Examples
Supercomputer	Fastest; massive parallel processing	Weather forecasting, research, simulation	PARAM (India), Cray, Fugaku
Mainframe	Very high; thousands of users at once	Banks, airlines, government, census	IBM Z series
Minicomputer	Medium; tens–hundreds of users	Department or mid-size business servers	PDP-11, midrange servers
Microcomputer	Single user; one microprocessor	Personal and everyday computing	Desktop, laptop, tablet, smartphone

Note: Microcomputers are by far the most common. Desktops, laptops, tablets, smartphones, servers, workstations, embedded systems and wearables are all microcomputers.

5. The block diagram of a computer

Whatever its type, every computer is built from the same **five functional units**, connected so that data and control signals can flow between them. This is the single most important diagram of the first year.

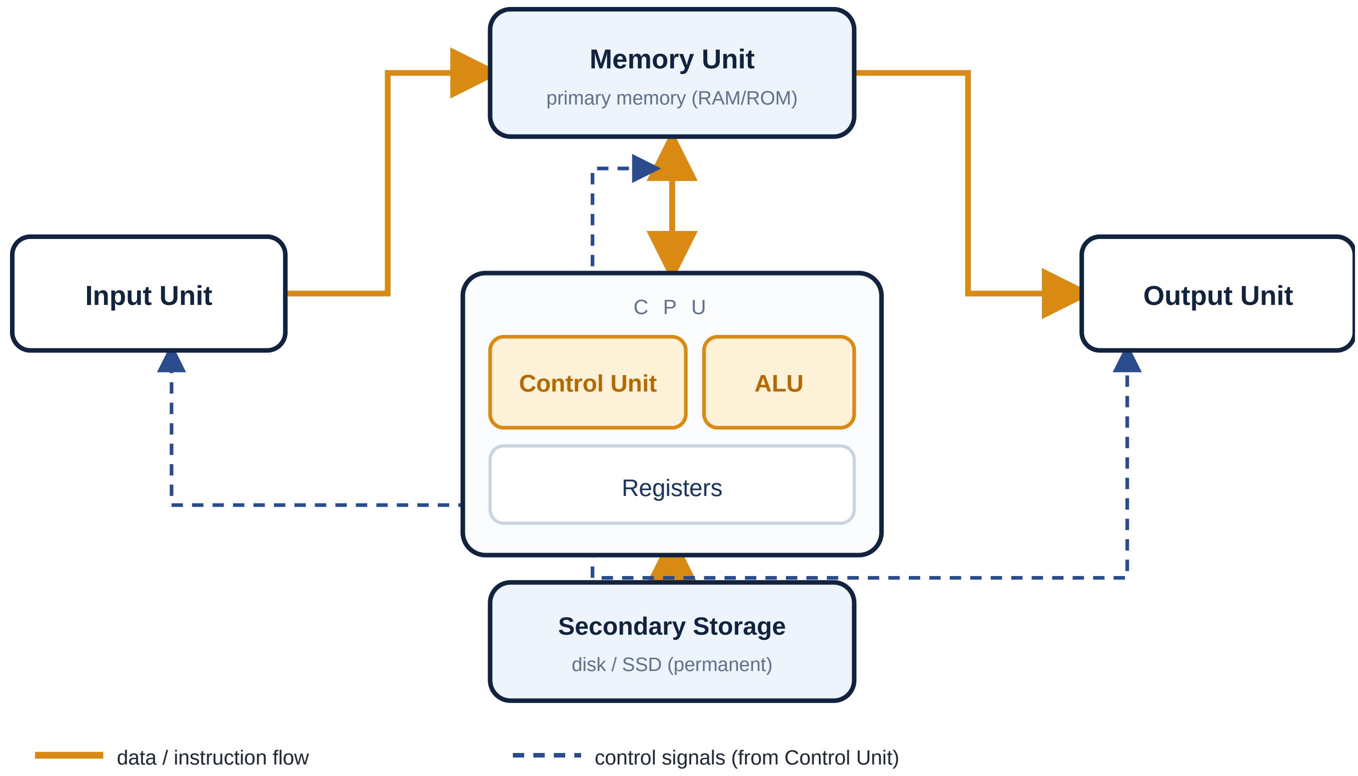


Fig 2. Block diagram of a computer. Solid arrows = data/instruction flow; dashed arrows = control signals from the Control Unit.

The functional units

- **Input Unit.** Accepts data and instructions from the outside world and converts them into binary form the computer can understand. Examples: keyboard, mouse, scanner, microphone.
- **Memory Unit (Primary/Main memory).** Temporarily stores data, instructions and intermediate results the CPU is currently using; it is fast and directly accessible by the CPU. It consists of **RAM** (temporary, read/write) and **ROM** (permanent start-up instructions).
- **Central Processing Unit (CPU).** The 'brain' that controls and processes. It contains:
  - **Control Unit (CU)** — fetches and decodes instructions and directs every other unit using control signals; it does not process data itself.
  - **Arithmetic Logic Unit (ALU)** — performs all arithmetic (add, subtract, etc.) and logic (compare, AND, OR) operations.
  - **Registers** — very small, very fast storage inside the CPU holding the values currently being worked on.
- **Output Unit.** Converts the binary results into human-readable form. Examples: monitor, printer, speaker.
- **Secondary Storage.** Permanent storage for programs and files, kept even when the power is off; larger, cheaper and slower than main memory and not accessed directly by the CPU. Examples: hard disk, SSD, pen drive.

Data flow versus control flow

- **Data / instruction flow** (solid arrows) carries the actual data and instructions between units — for example Input → Memory → CPU → Memory → Output. These travel on the **data bus**.

- **Control flow** (dashed arrows) carries the control signals sent out by the Control Unit to coordinate and time every unit. These travel on the **control bus**.

> **Chapter summary.** Classify by data (analog/digital/hybrid), purpose (general/special) and size (super/mainframe/mini/micro). Every computer follows one block diagram: Input → Memory → CPU (CU + ALU) → Output, with Secondary Storage; the CU controls all units while the ALU processes data.

2-Mark Questions (Short Answer)

Answer in 2–4 lines.

Q1. On what three bases are computers classified? 2 marks

By the data they handle (analog, digital, hybrid); by purpose (general- and special-purpose); and by size/power (super, mainframe, mini, micro).

Q2. What is an analog computer? Give one example. 2 marks

An analog computer works with continuous physical quantities which it measures, giving approximate results. Example: a speedometer or mercury thermometer.

Q3. Differentiate between digital and hybrid computers. 2 marks

A digital computer works only with discrete data (0s and 1s) and counts exactly. A hybrid computer takes analog measurements as input but processes them digitally, e.g. an ICU patient monitor.

Q4. Define general-purpose and special-purpose computers. 2 marks

A general-purpose computer can do many different tasks by changing its program (e.g. a laptop). A special-purpose computer is built for one specific task only (e.g. a washing-machine controller).

Q5. What is a supercomputer? Give one Indian example. 2 marks

A supercomputer is the fastest and most powerful class, using massive parallel processing for complex tasks like weather forecasting. Indian example: PARAM.

Q6. Why are mainframe computers used by banks and airlines? 2 marks

Because they are very reliable and can serve thousands of users at the same time, which such large organisations need.

Q7. Name the five functional units of a computer. 2 marks

Input Unit, Memory Unit, Control Unit, Arithmetic Logic Unit (ALU) and Output Unit (with Secondary Storage supporting them).

Q8. What is the function of the Control Unit? 2 marks

The Control Unit fetches and decodes instructions and directs all other units using control signals; it manages operations but does not process data itself.

Q9. What is the difference between primary and secondary memory? 2 marks

Primary memory (RAM/ROM) is fast, directly accessed by the CPU and mostly temporary; secondary storage (disk/SSD) is slower, larger, cheaper and permanent.

Q10. What is the difference between data flow and control flow in the block diagram? 2 marks

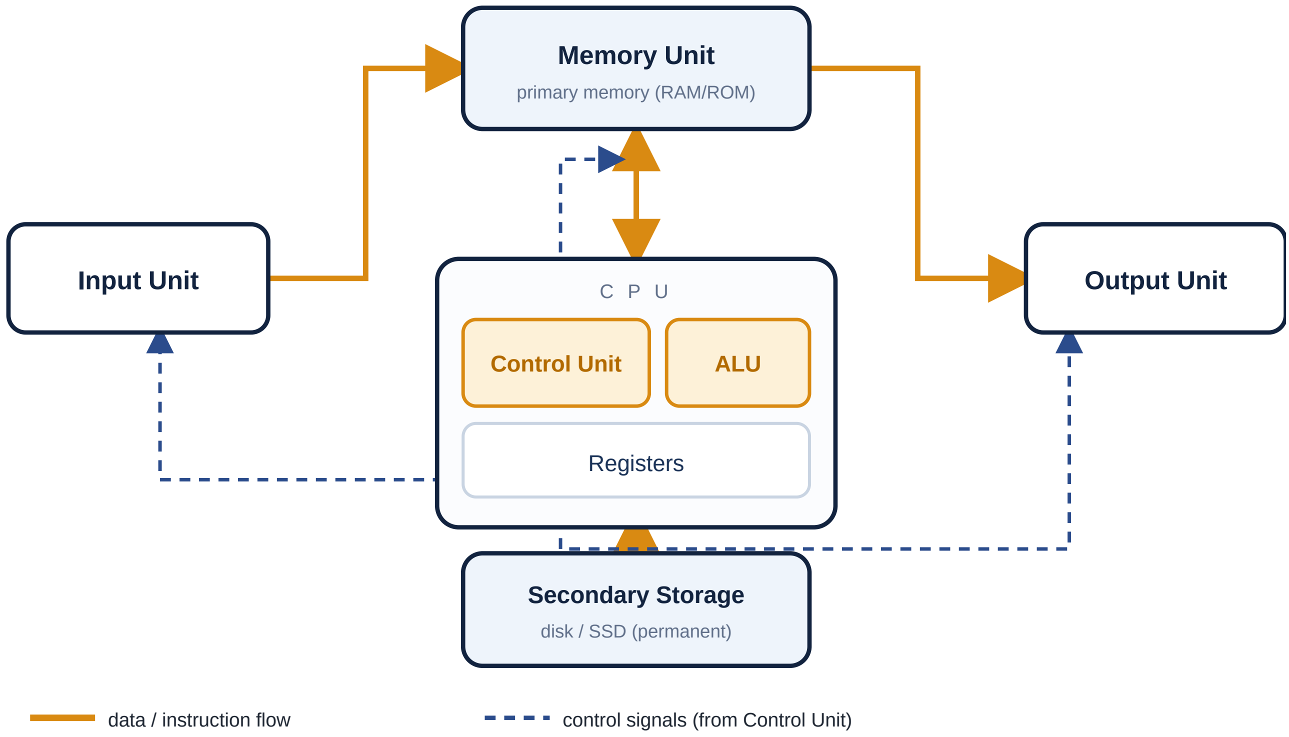
Data flow carries the actual data and instructions between units (data bus); control flow carries the Control Unit's timing/coordination signals (control bus).

14-Mark Questions (Long Answer)

For the block-diagram question you must include the labelled diagram, explain every unit, and describe data and control flow.

Q1. Draw the block diagram of a computer and explain the function of each unit in detail. 14 marks

Introduction. Every computer, from a supercomputer to a smartphone, is built from the same five functional units connected by buses. The block diagram shows these units and how information moves between them.



Block diagram of a computer. Solid = data flow; dashed = control signals.

The functional units.

- **Input Unit.** Accepts data and instructions from the user and converts them into the binary form the computer understands. Examples: keyboard, mouse, scanner.
- **Memory Unit (primary memory).** Temporarily holds the data, instructions and intermediate results the CPU is using. It is fast and directly accessible by the CPU, and is made of RAM (temporary) and ROM (permanent start-up).
- **Central Processing Unit (CPU).** The brain of the computer. It contains the *Control Unit*, which fetches, decodes and directs operations with control signals; the *Arithmetic Logic Unit (ALU)*, which performs all calculations and comparisons; and *registers*, tiny fast stores for current values.
- **Output Unit.** Converts the binary results back into human-readable form. Examples: monitor, printer, speaker.
- **Secondary Storage.** Permanent storage for programs and files, retained when the power is off; larger, cheaper and slower than main memory. Examples: hard disk, SSD.

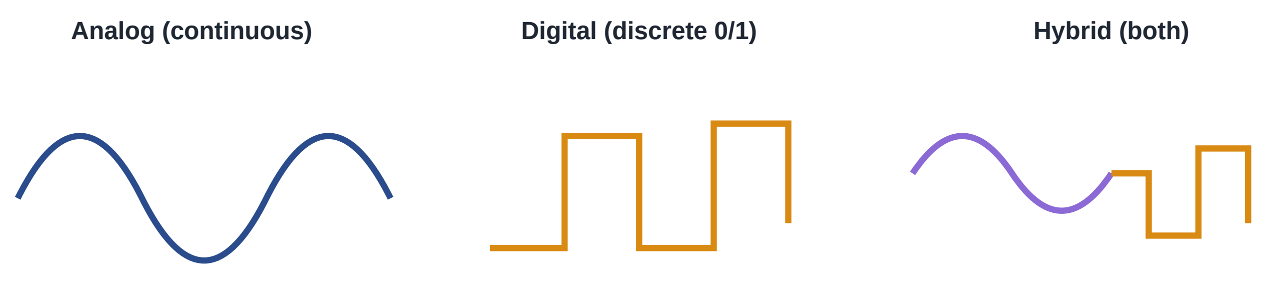
How it works together. Input sends data to memory; the CPU fetches it, the ALU processes it under the Control Unit's direction, the result returns to memory, and finally goes to the output unit — while the Control Unit coordinates every step.

Conclusion. The block diagram is the universal plan of a computer: five units working together, with the CPU at the centre controlling and processing all information.

Q2. Explain the classification of computers on the basis of the data they handle and on the basis of size. 14 marks

Introduction. Computers can be grouped in several ways. Two important bases are the kind of data they handle and their size and power.

A. By the data handled.



Continuous (analog), discrete (digital) and combined (hybrid) signals.

- **Analog computers** work with continuous physical quantities such as voltage or temperature, which they measure. Results are approximate. Examples: speedometer, mercury thermometer, seismograph.
- **Digital computers** work with discrete data — the binary digits 0 and 1 — which they count, giving exact results. Almost all modern computers are digital. Examples: PC, laptop, smartphone.
- **Hybrid computers** combine both: they take analog measurements as input and process them digitally, giving speed with accuracy. Example: an ICU patient monitor or a petrol pump.

B. By size and power.

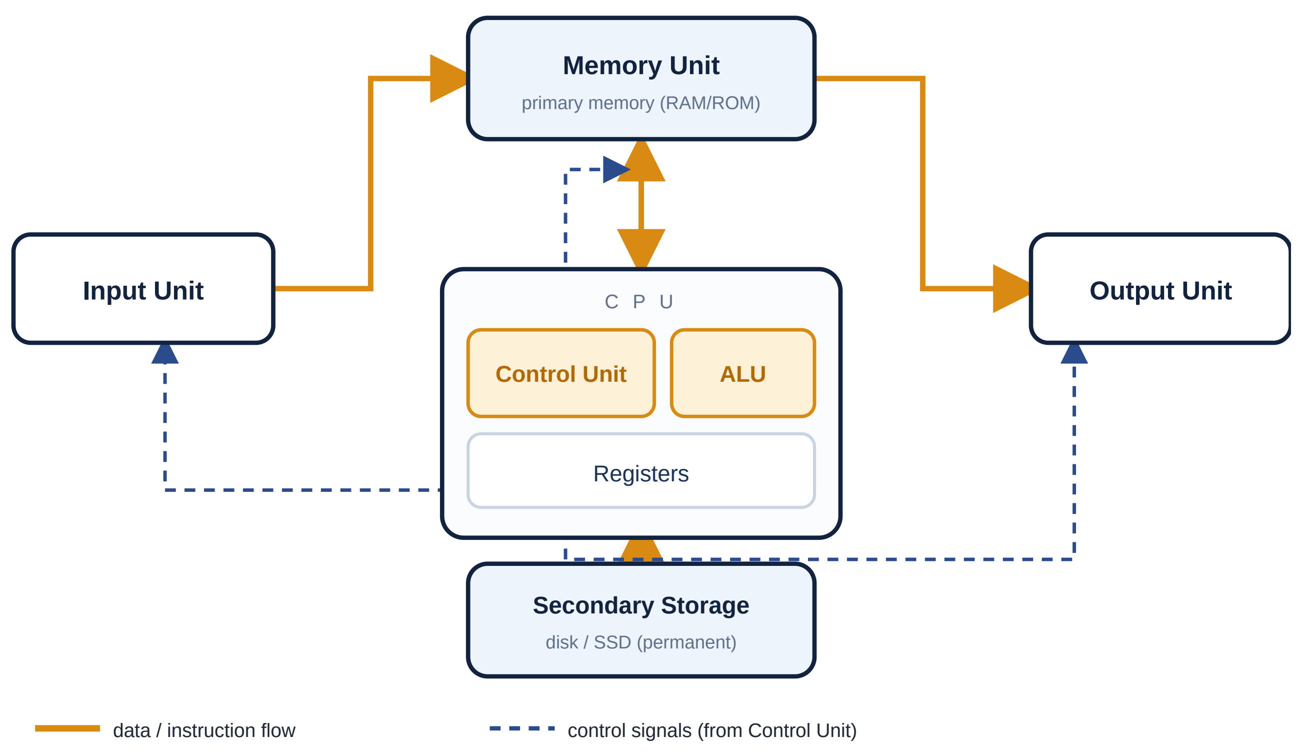
Type	Users / power	Use	Examples
Supercomputer	Fastest; massive parallel	Weather, research, simulation	PARAM, Cray
Mainframe	Thousands of users; very reliable	Banks, airlines, government	IBM Z series
Minicomputer	Tens–hundreds of users	Departmental / mid-size business	PDP-11, midrange servers
Microcomputer	Single user; one microprocessor	Personal computing	Desktop, laptop, smartphone

Conclusion. Classifying by data separates measuring from counting machines, while classifying by size ranges from nation-scale supercomputers down to the single-user microcomputers we use every day.

Q3. What is the CPU? Explain its parts and describe how it works together with memory, input and output units. **14 marks**

The CPU. The Central Processing Unit is the brain of the computer. It controls all activities and carries out all processing. It has three main parts.

- **Control Unit (CU).** Directs the operation of the whole computer. It fetches instructions from memory, decodes them, and sends control signals to every unit to tell it what to do and when. It does not process the data itself — it is like a conductor.
- **Arithmetic Logic Unit (ALU).** Performs all arithmetic operations (addition, subtraction, multiplication, division) and logic operations (comparison, AND, OR, NOT). It is built from logic gates.
- **Registers.** Small, extremely fast storage locations inside the CPU that hold the data and instructions currently being processed.



The CPU (with CU and ALU) at the centre of the block diagram.

Working with the other units. The **input unit** supplies data to memory; the CPU **fetches** the instruction and data from **memory**, the ALU **processes** them under the Control Unit's direction, and the result is stored back in memory and passed to the **output unit**. Secondary storage keeps programs and results permanently. Throughout, control signals from the CU coordinate the timing of every unit.

Conclusion. The CPU, through its Control Unit and ALU, both directs and performs all processing, working hand in hand with memory and the input/output units to turn data into useful information.